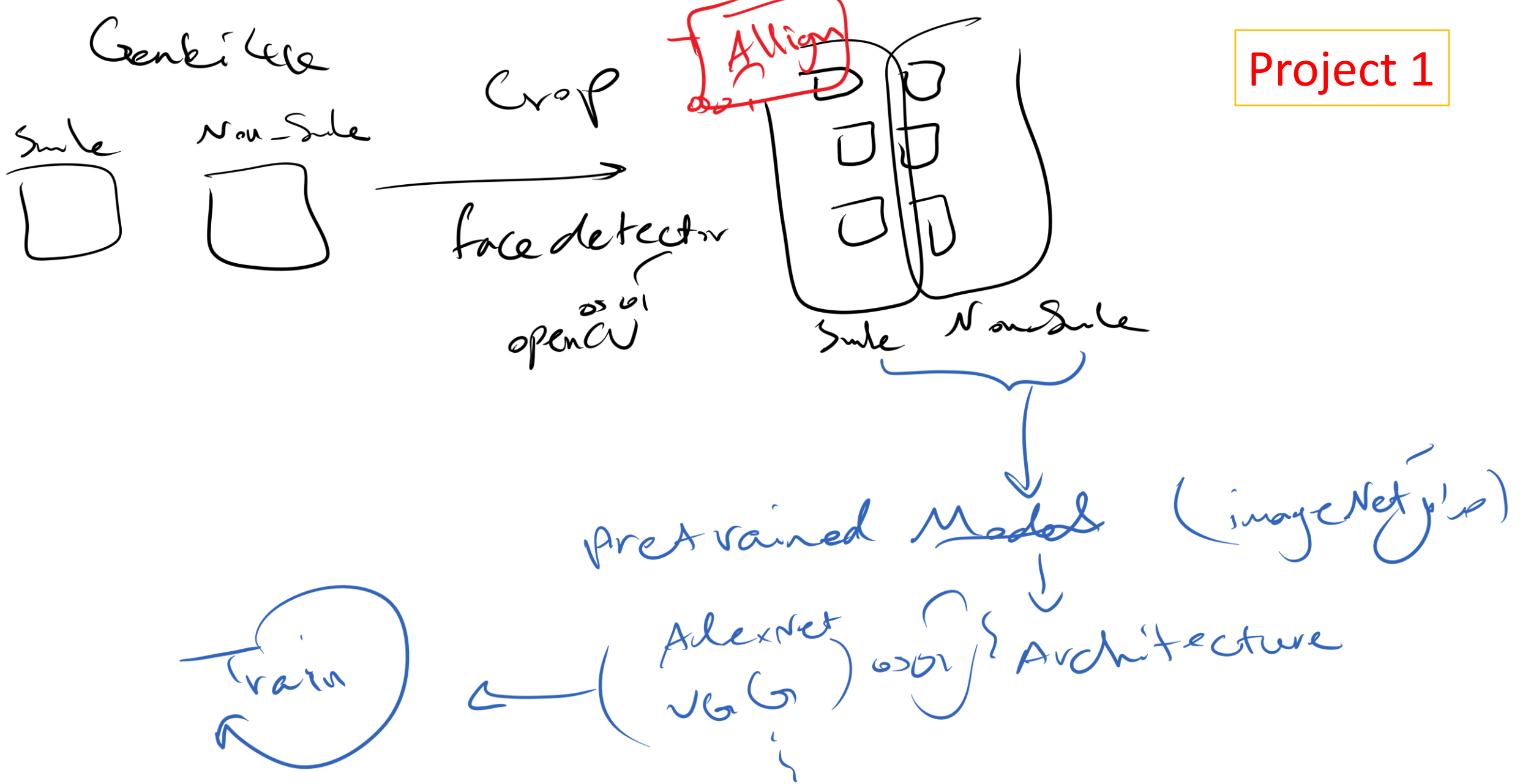
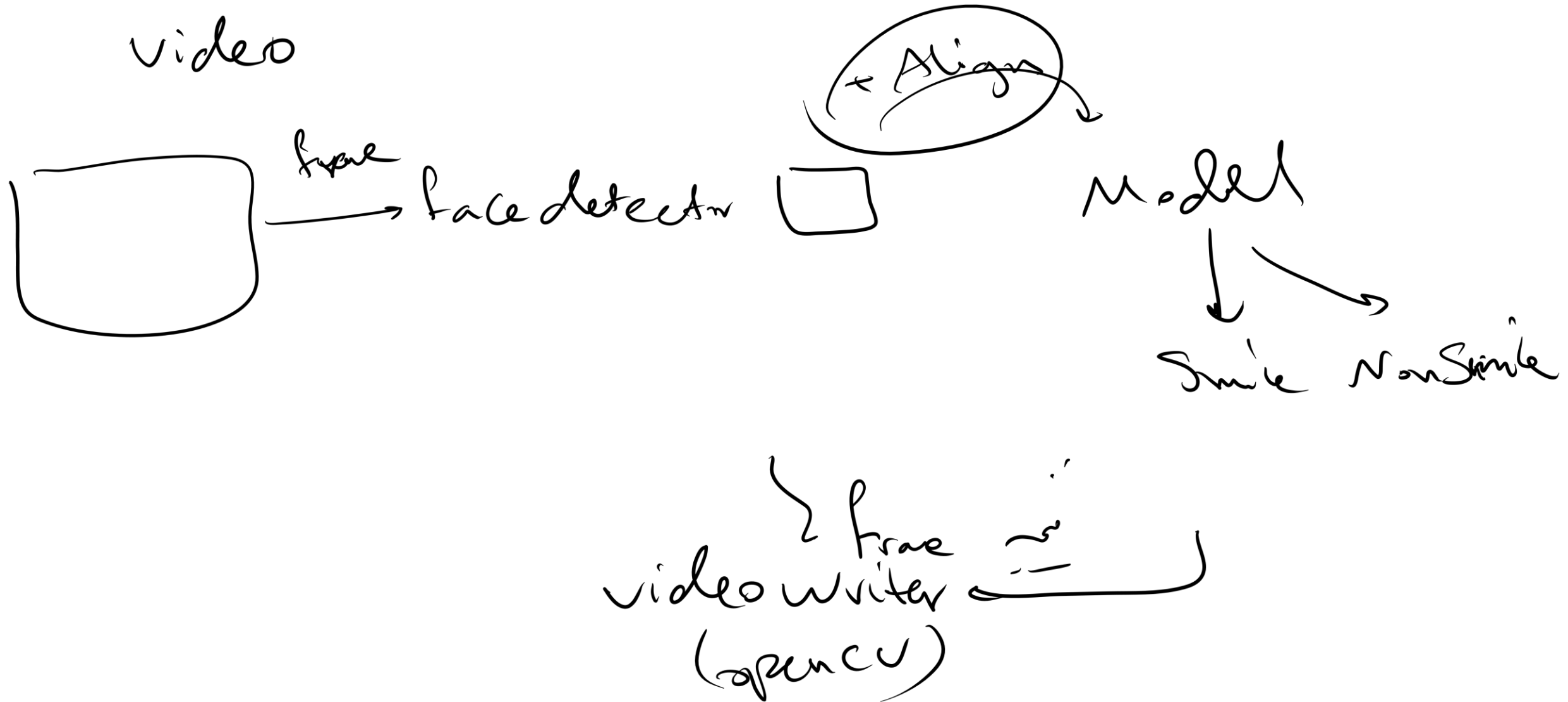
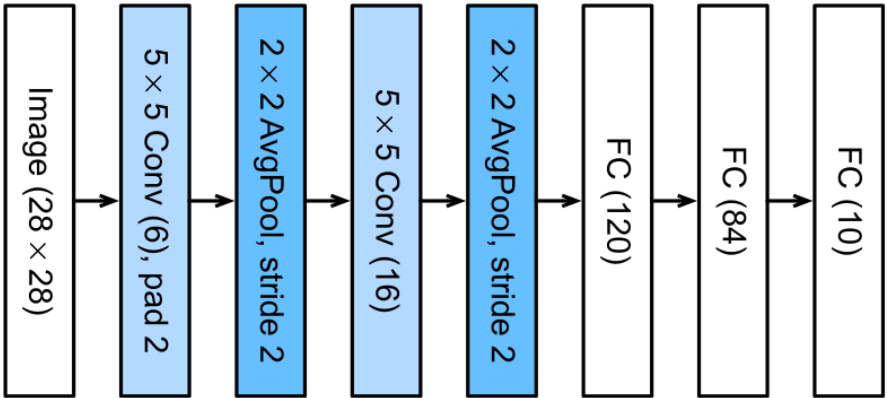






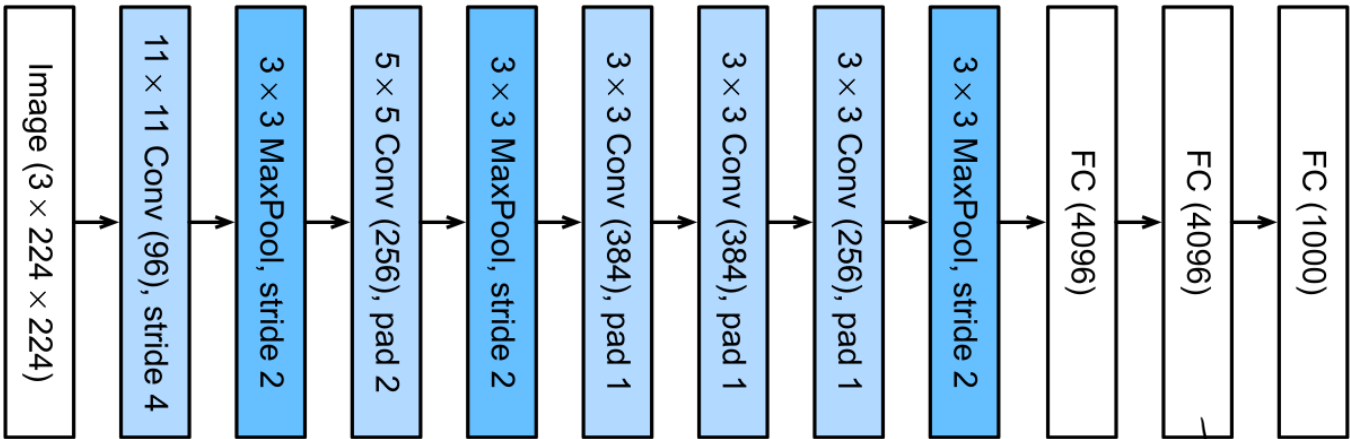
lenNet

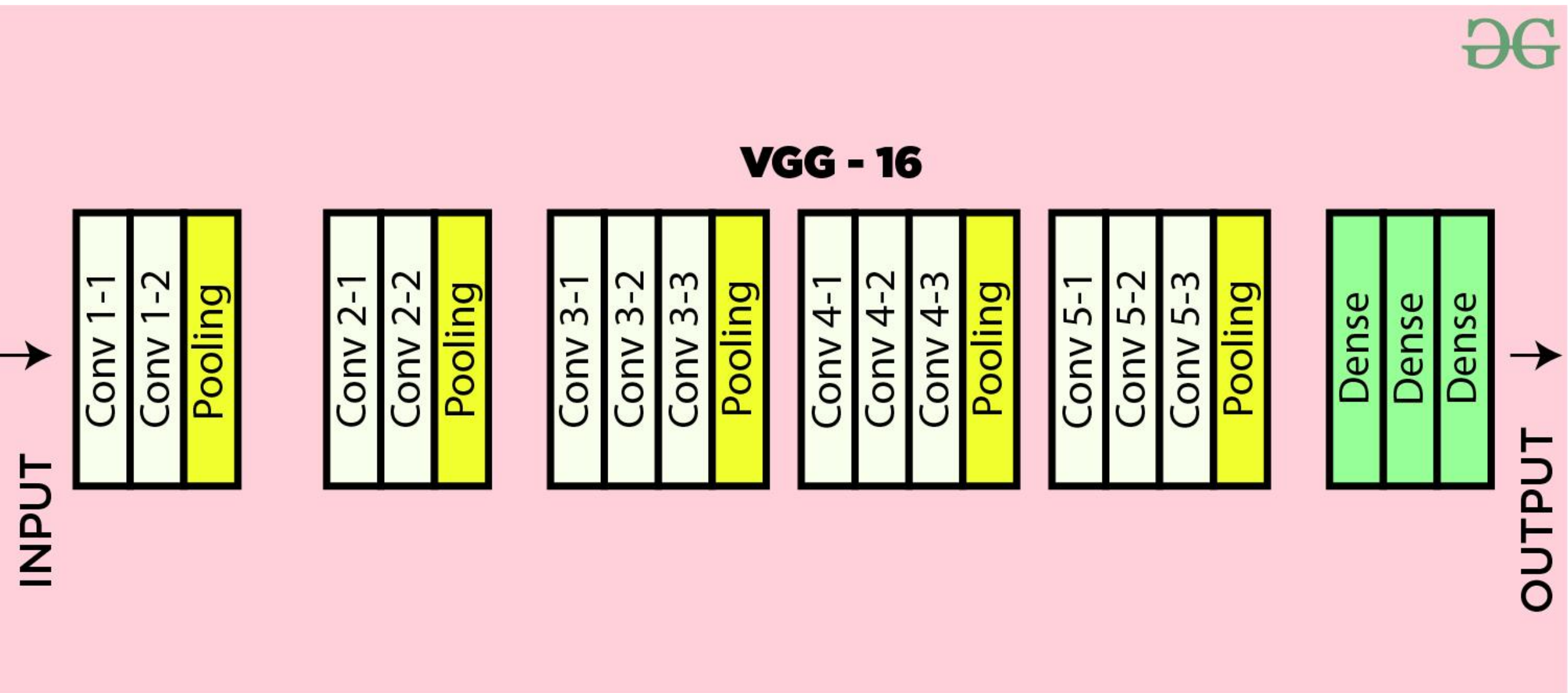


AlexNet

2012

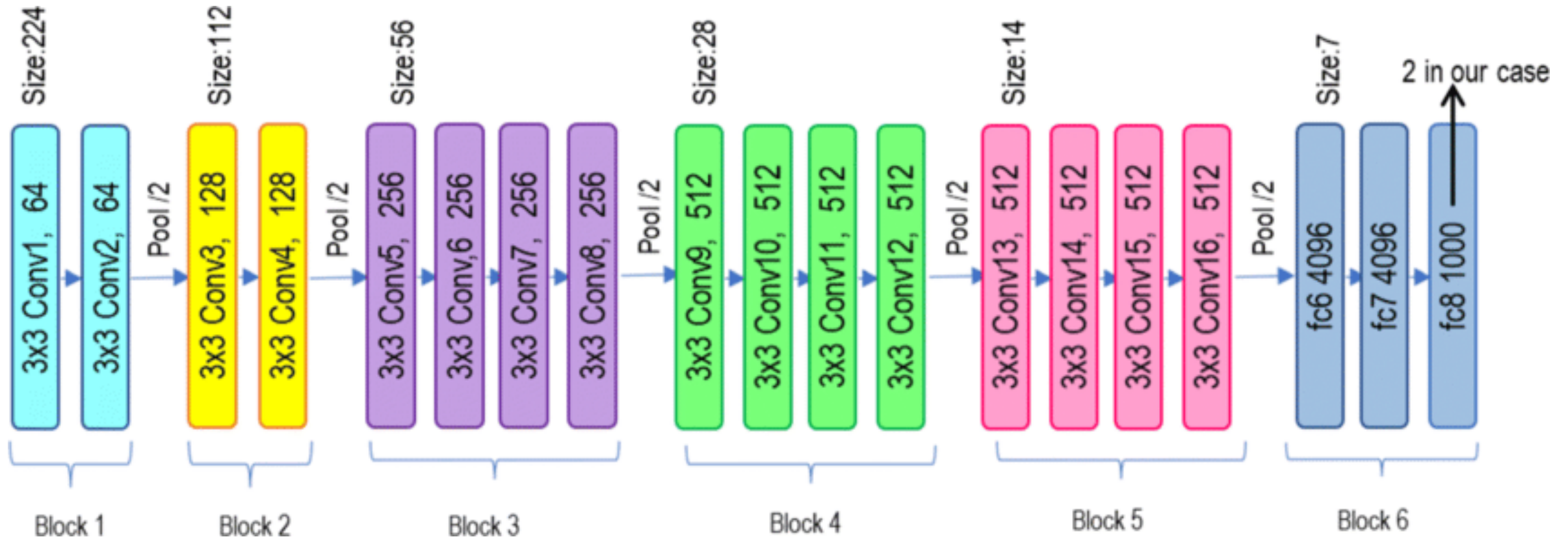
zfNet
2013



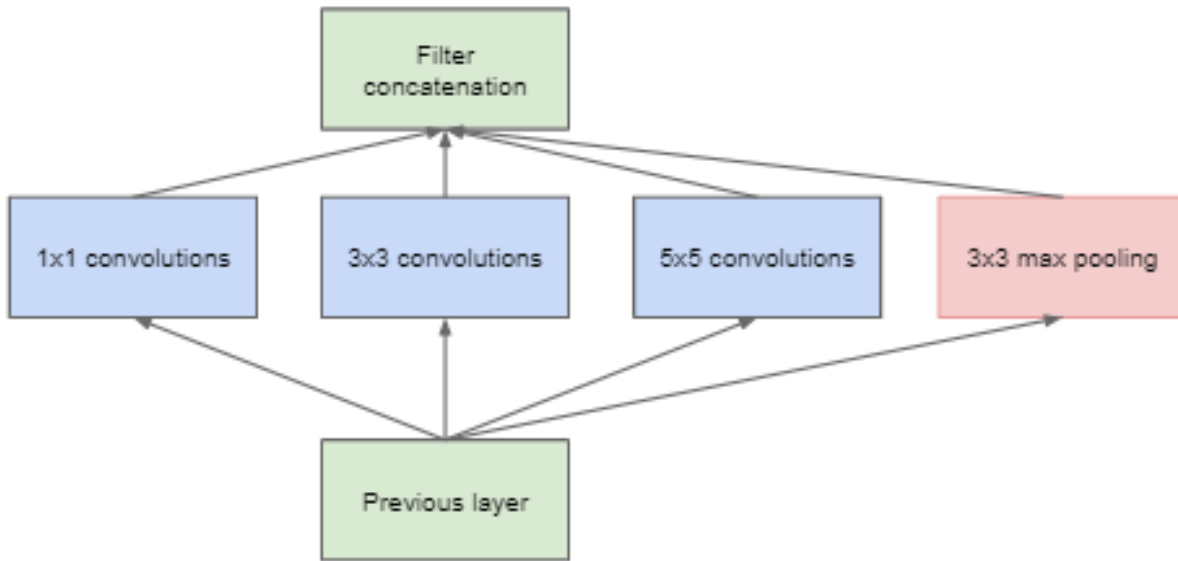


VGG Architecture

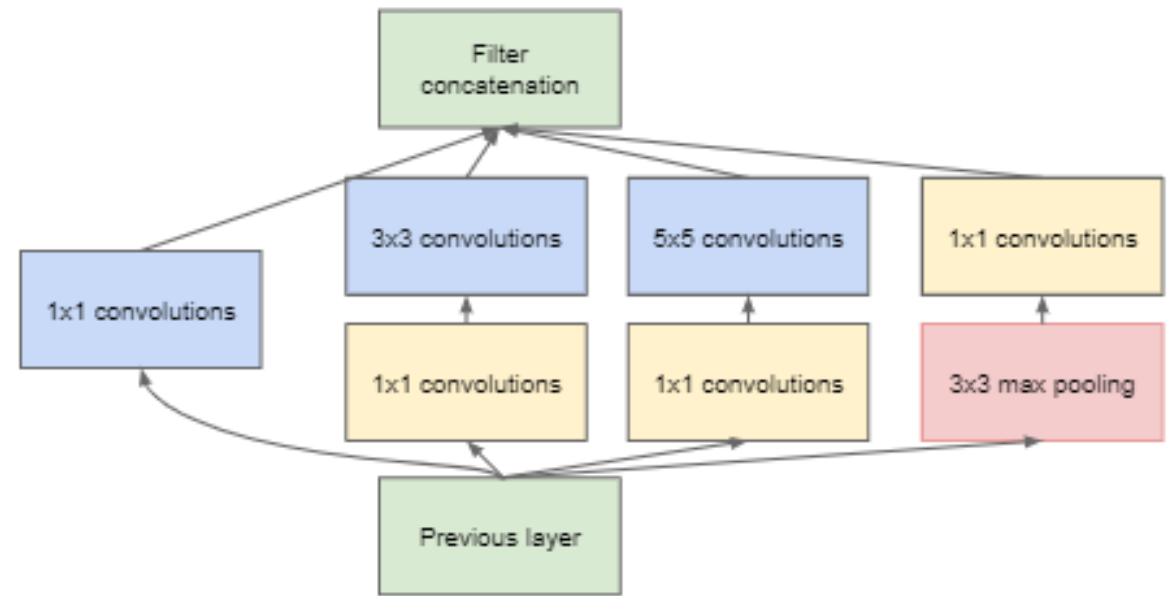
(2014)



Inception Module

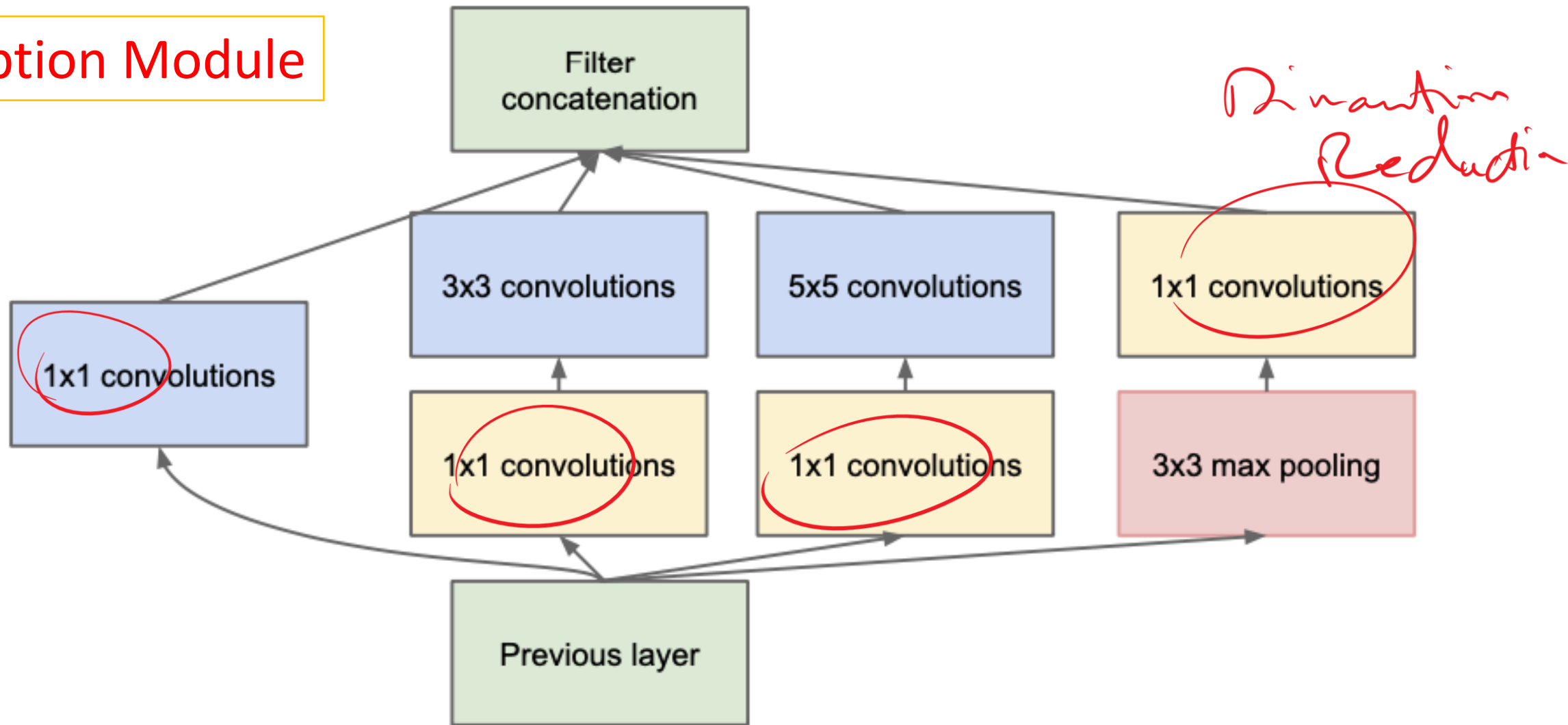


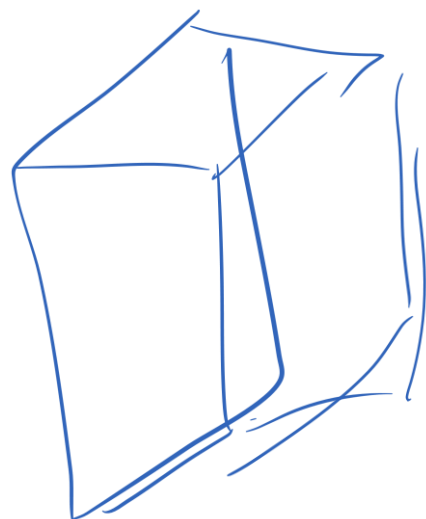
(a) Inception module, naïve version



(b) Inception module with dimension reductions

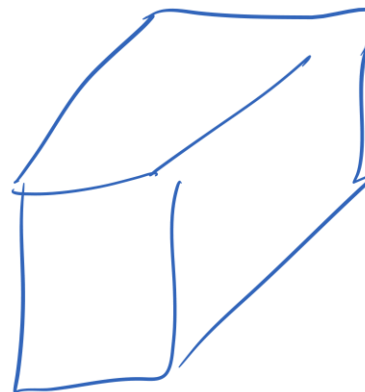
Inception Module





$640 \times 640 \times 64$

Conv 1×1
 $\xrightarrow{\times 32}$
filter $\uparrow$



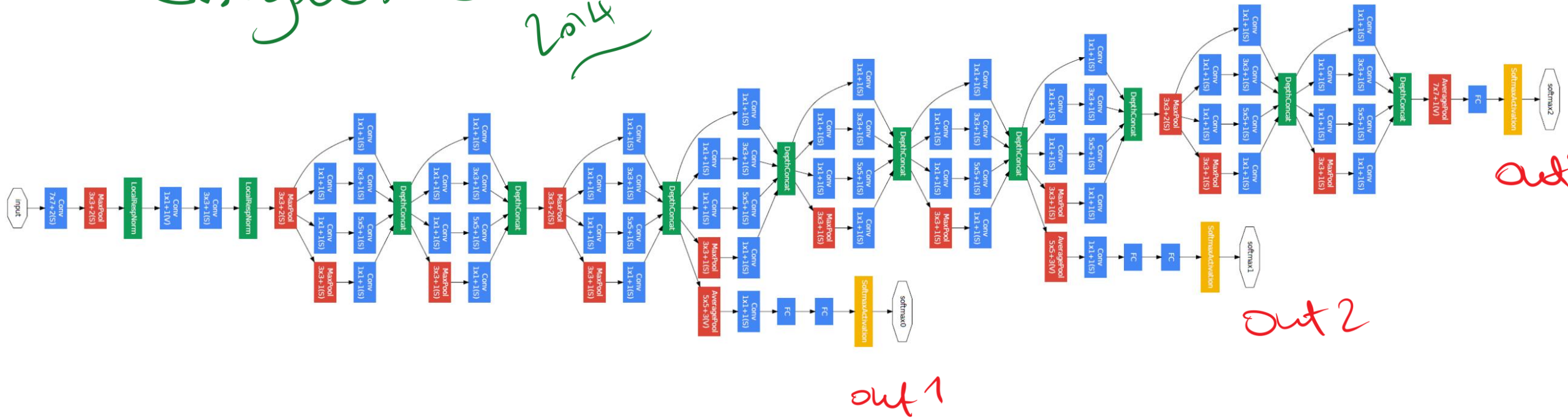
$640 \times 640 \times 32$



GoogleNet Architecture

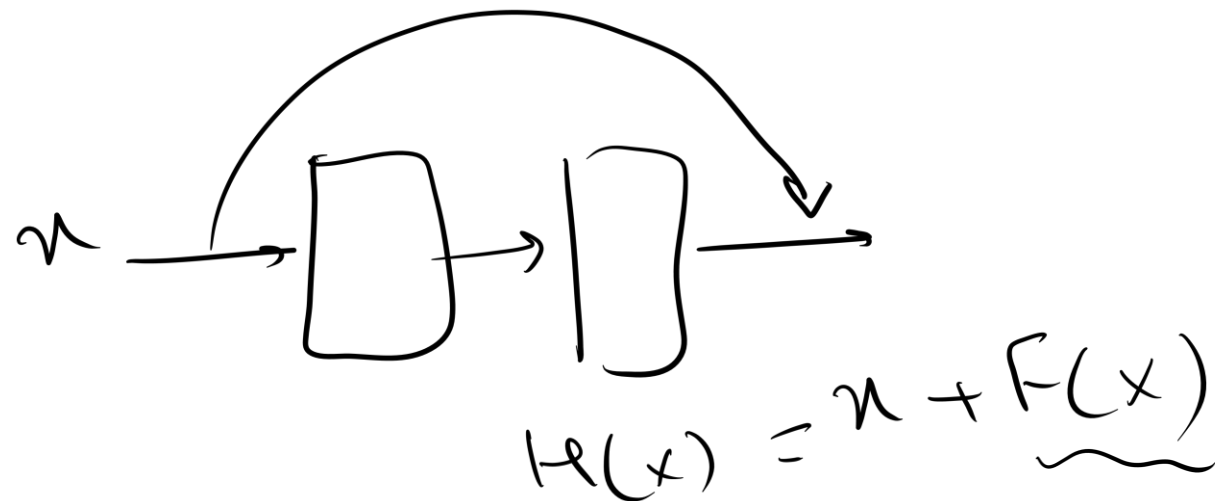
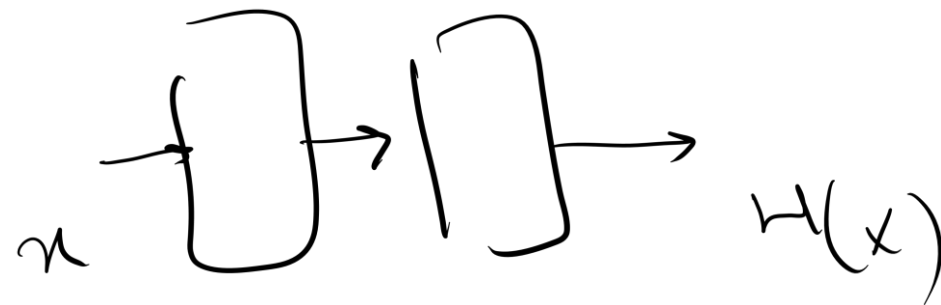
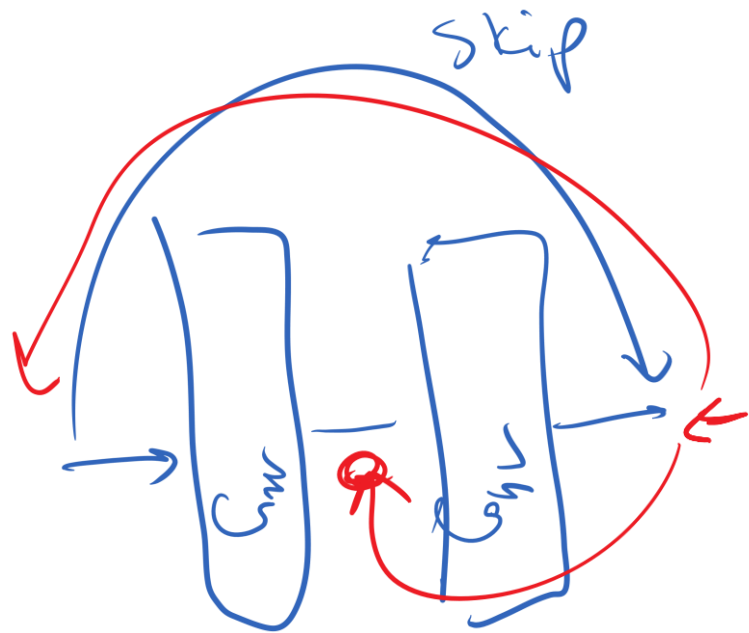
Inception Modules

GoogleNet 2014



ResNe
2014

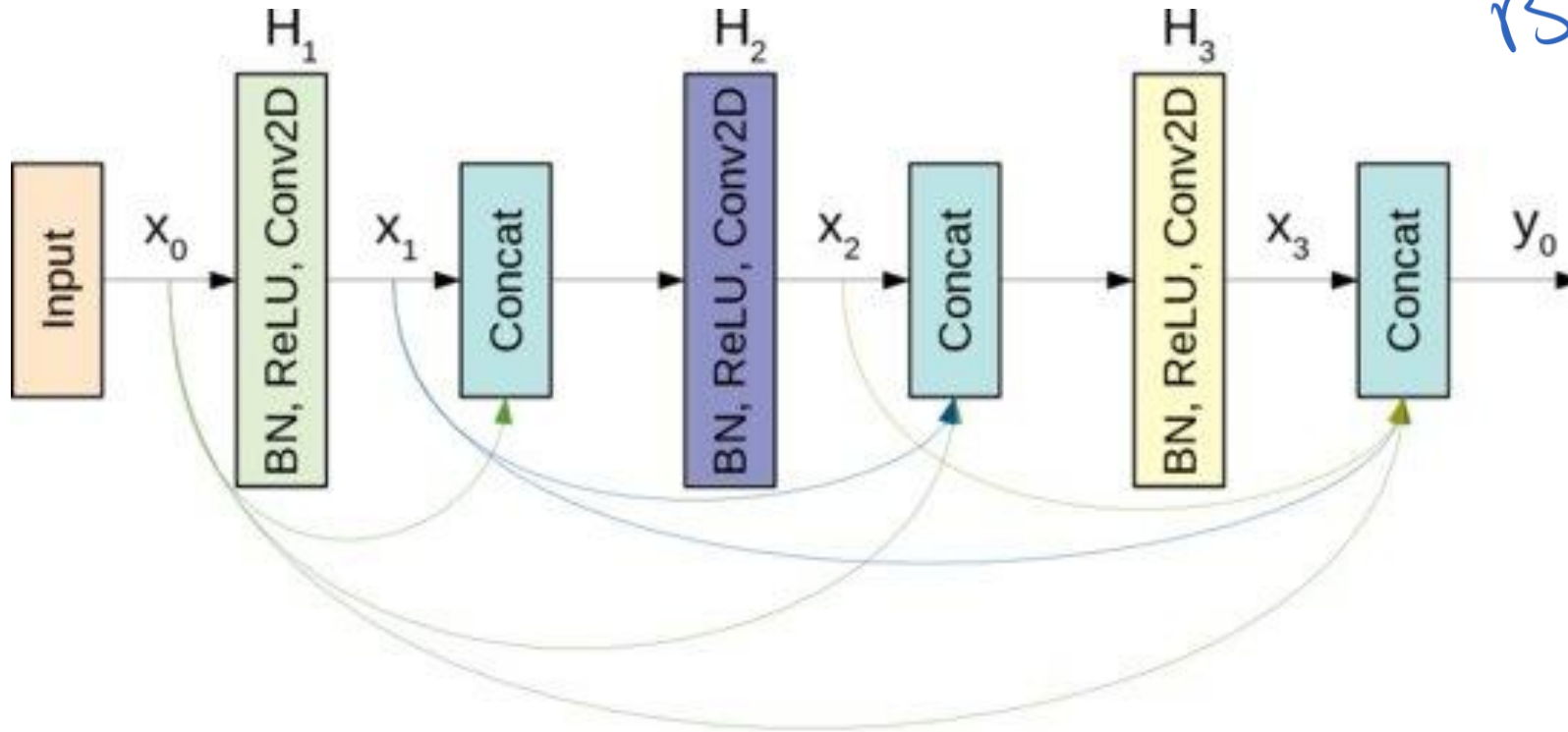




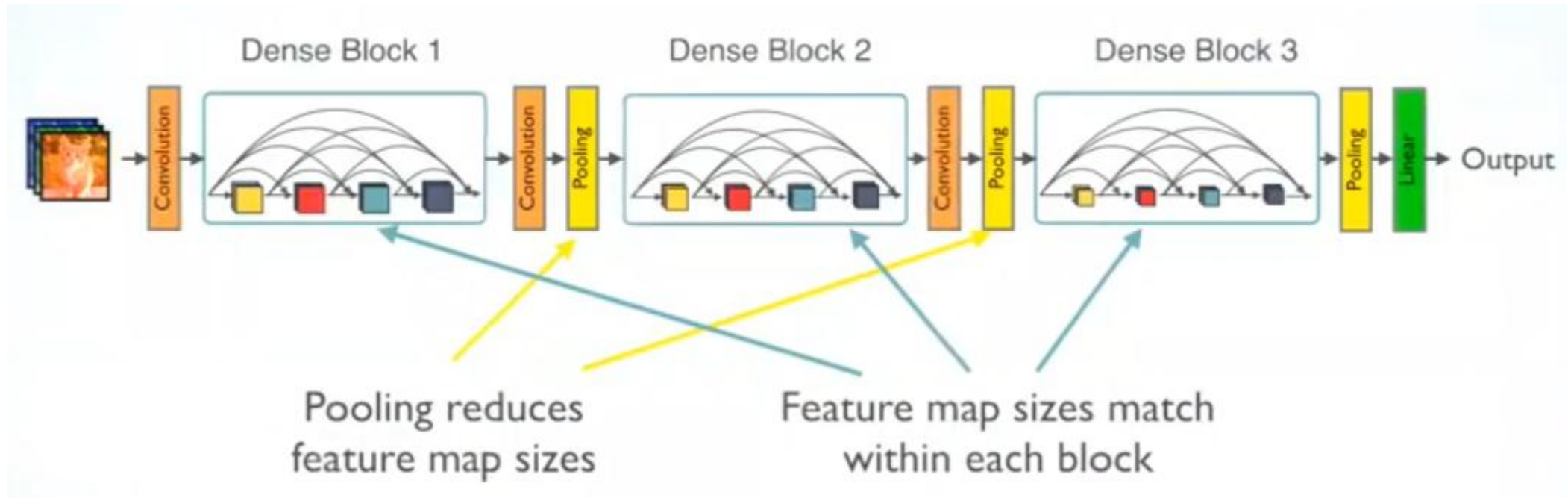
DenseNet 2017

Dense Module

Dense
Block



DenseNet Architecture

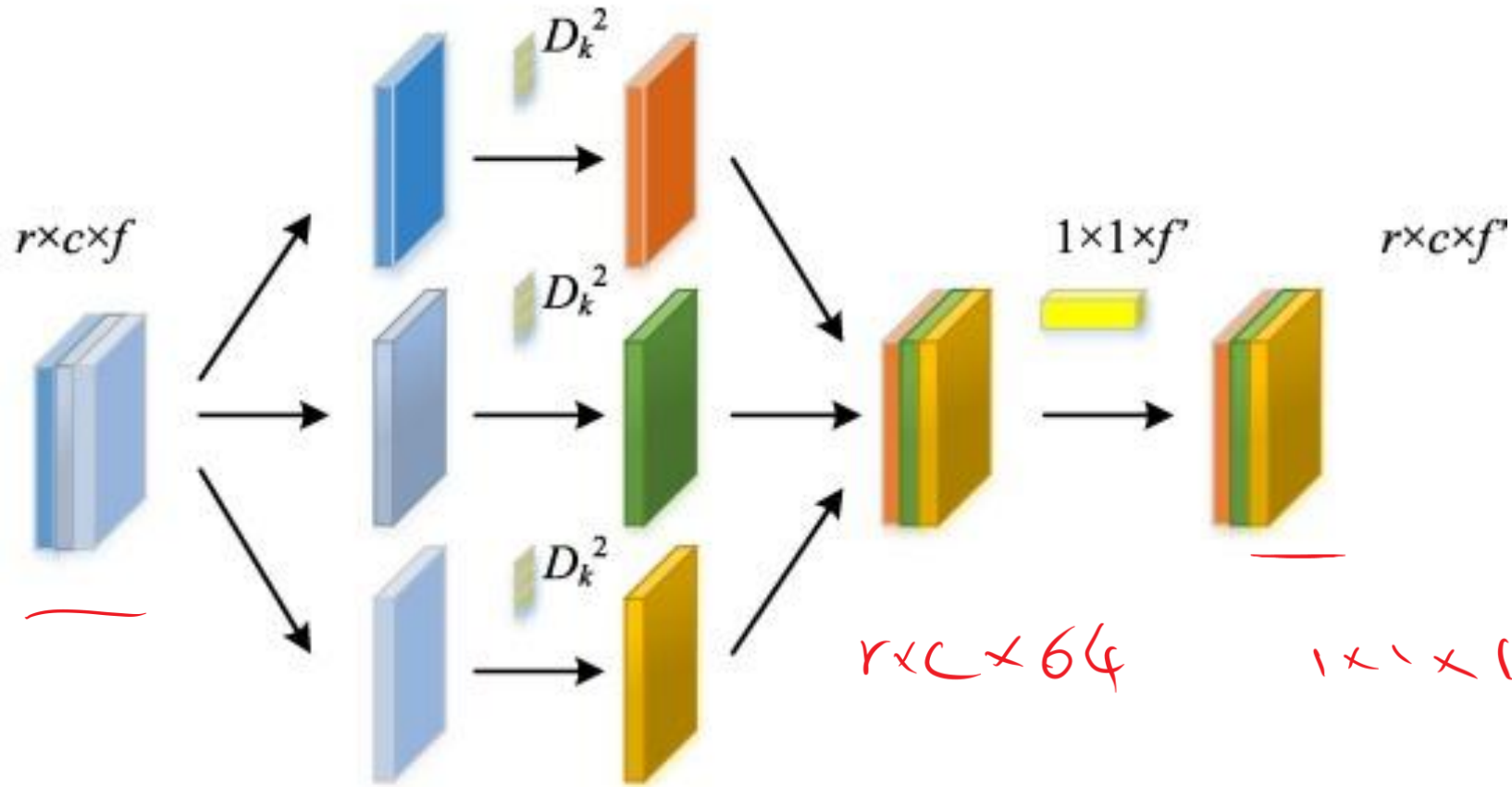


Depthwise
Separable
Conv.

MobileNet Module

Depth Separable Convolution (DSC)
 $D_k \times D_k, f, f'$

MobileNet
2017



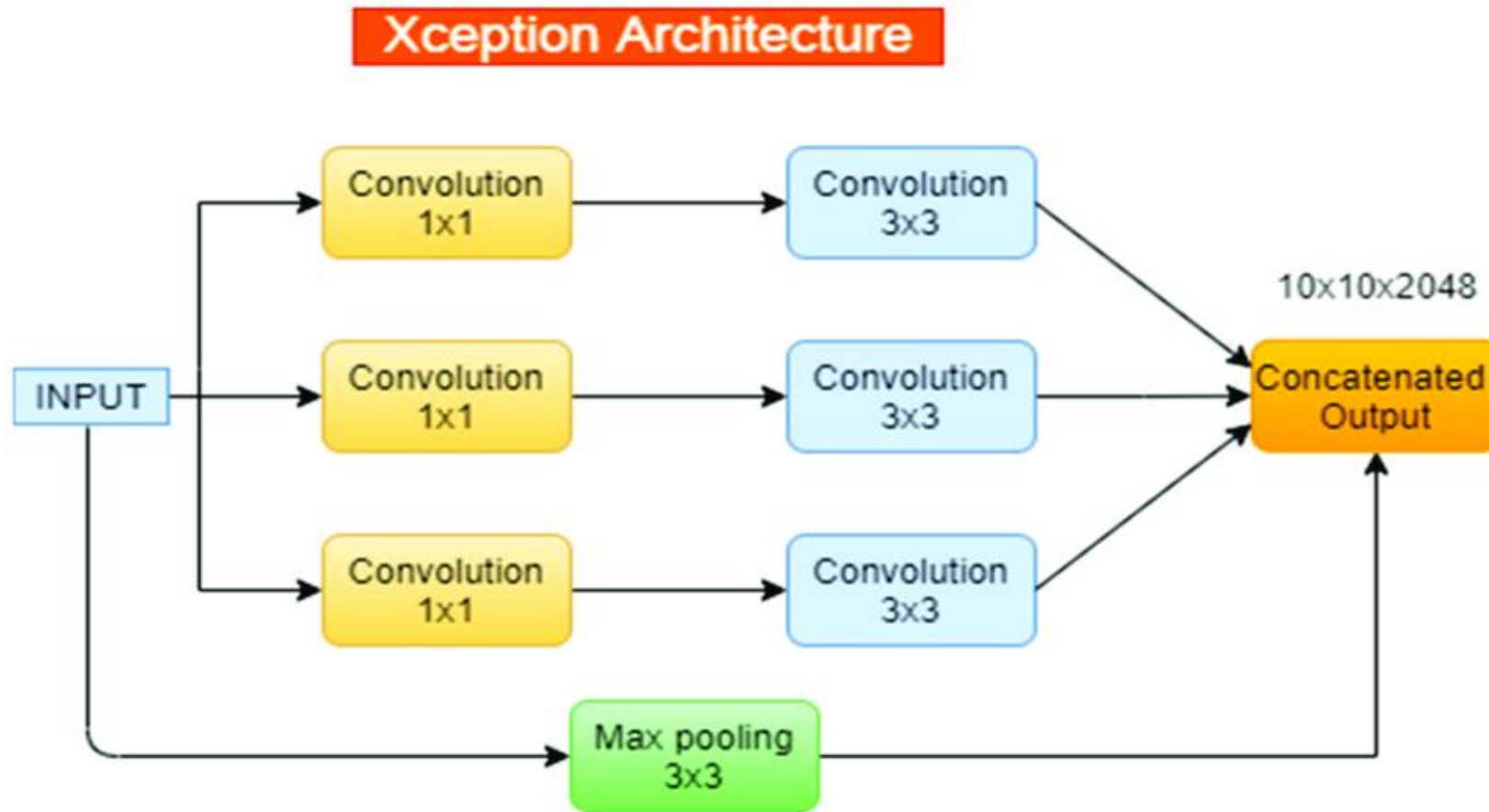
$r \times c \times 64$

$1 \times 1 \times 16$

$\rightarrow r \times c \times 16$

Xception 2017

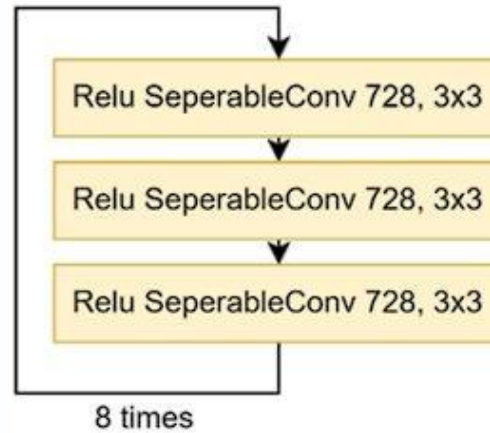
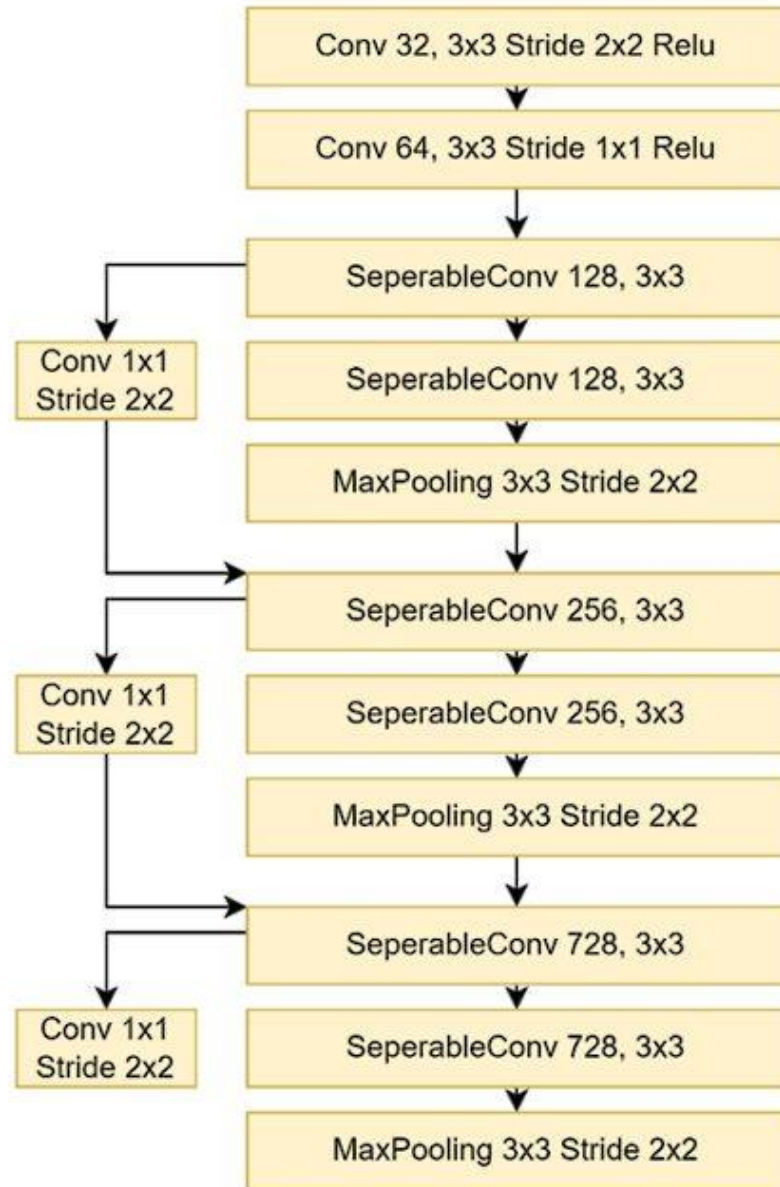
Xception Module



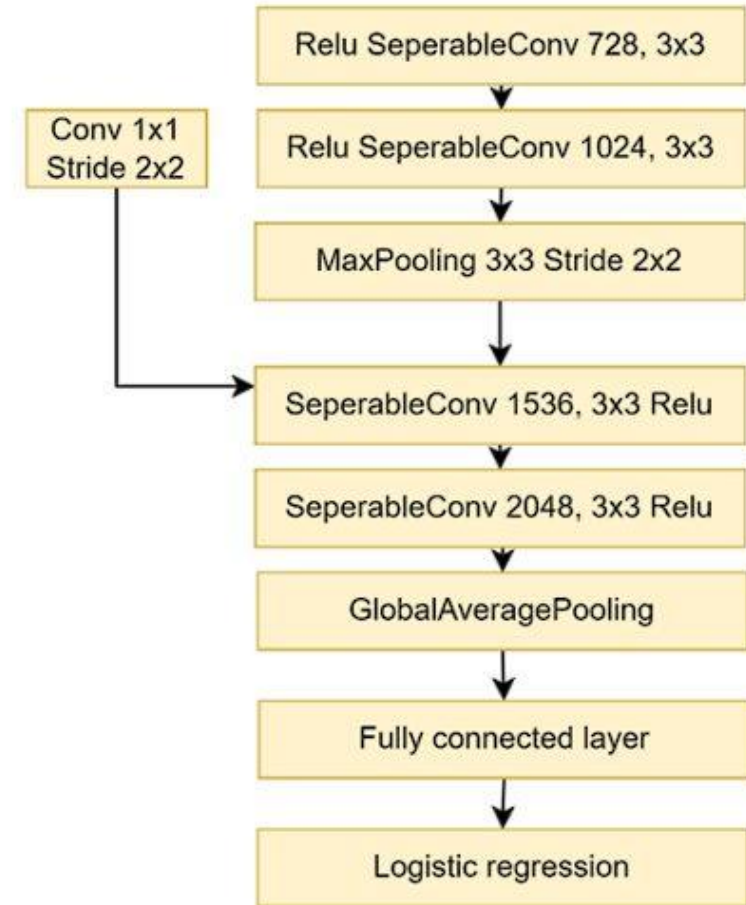
Entry flow

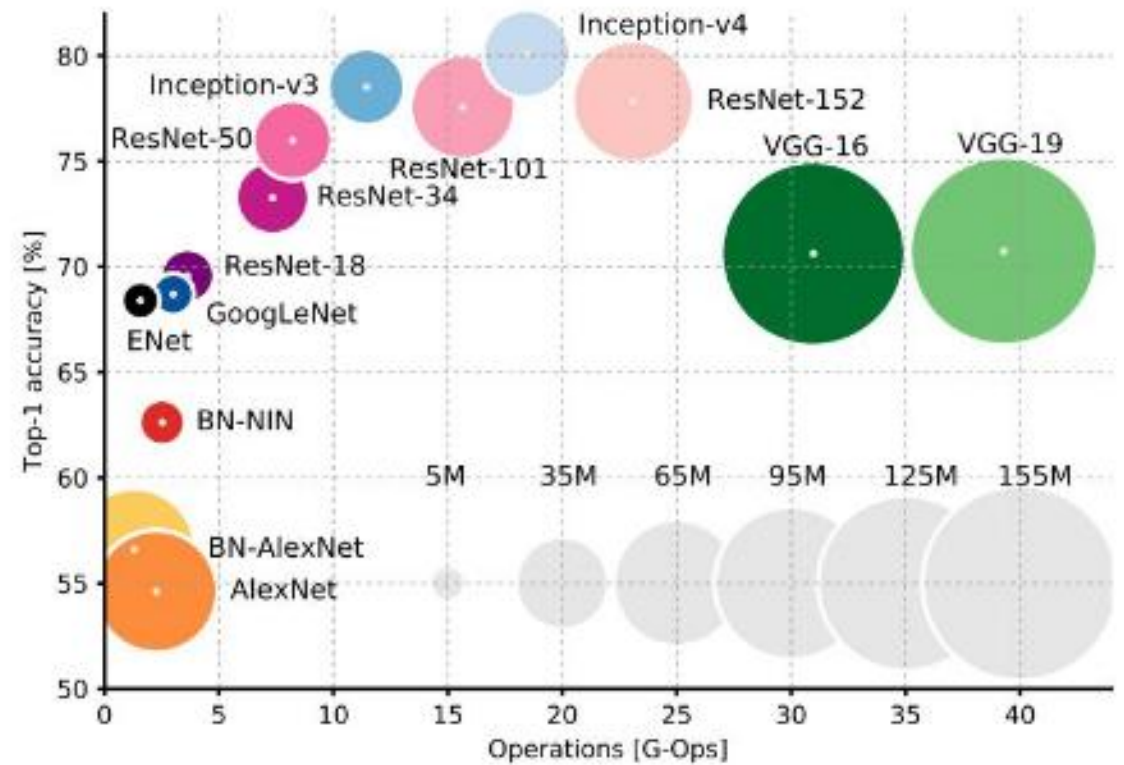
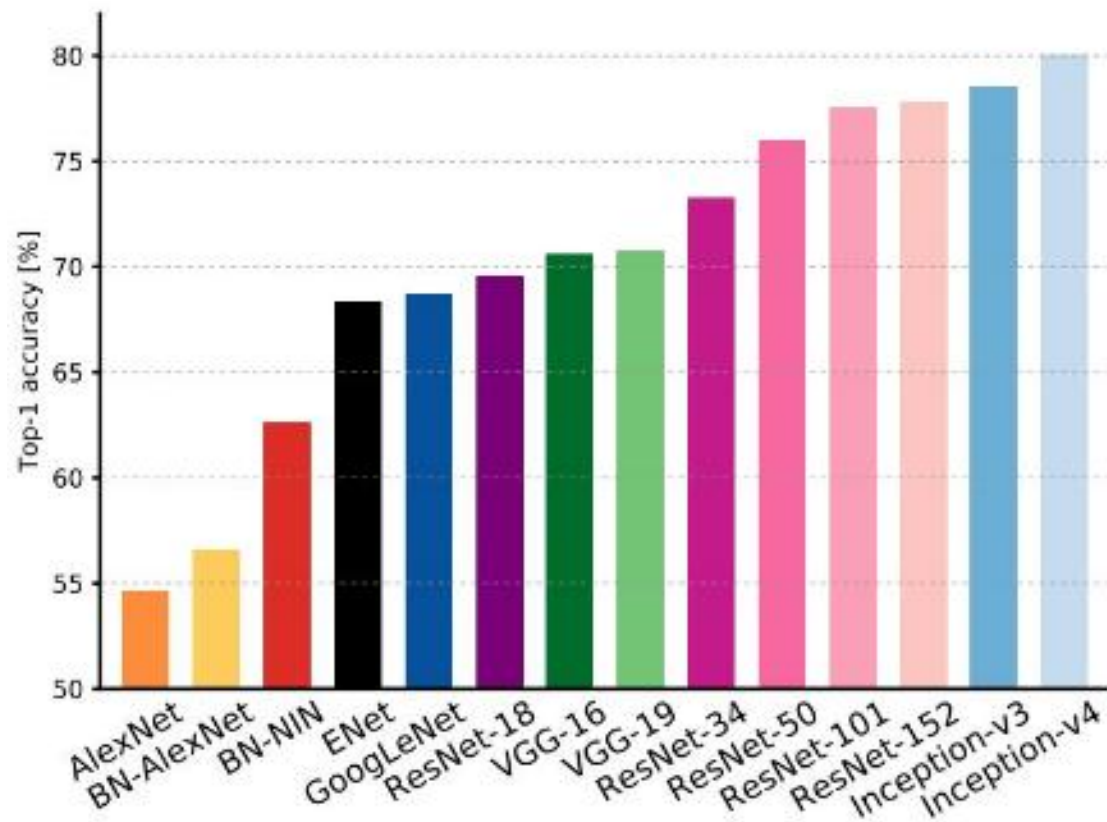
Middle flow

Exit flow



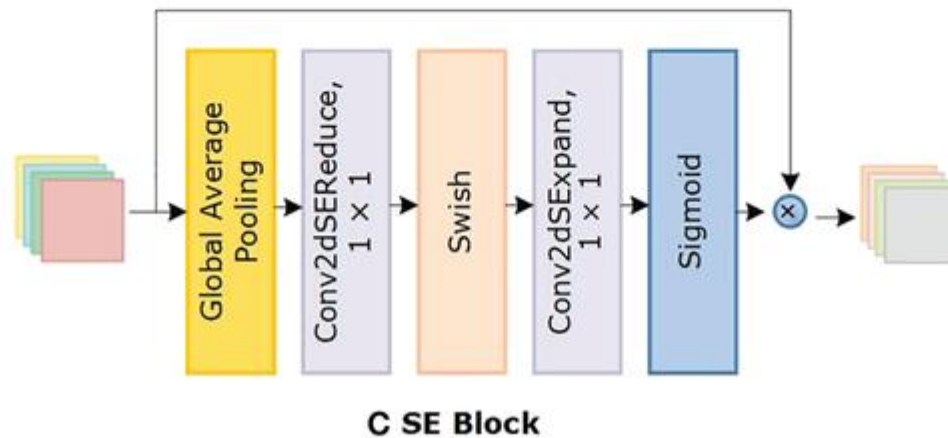
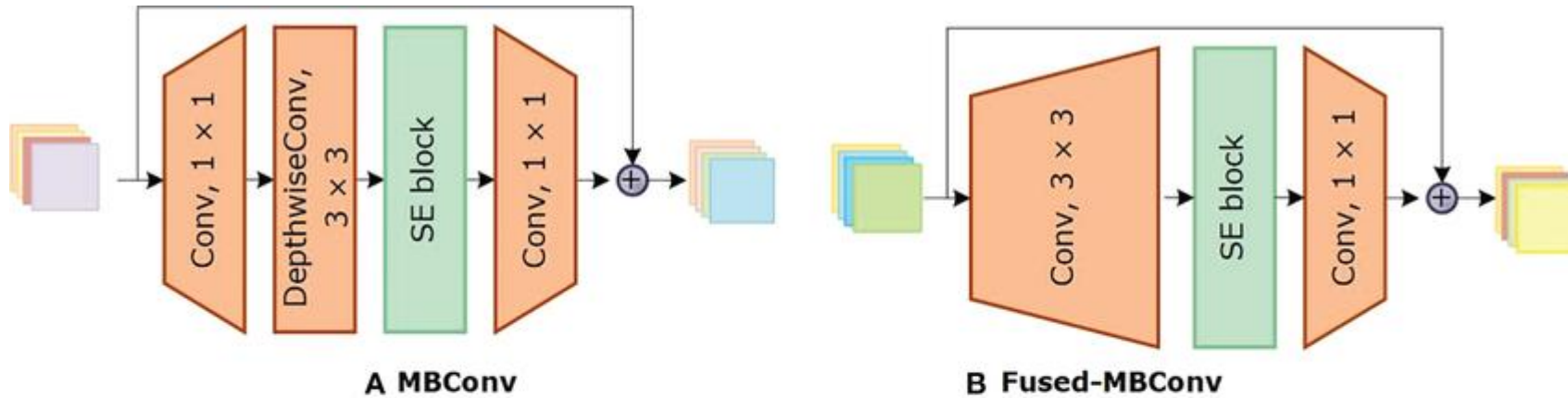
Xception Architecture





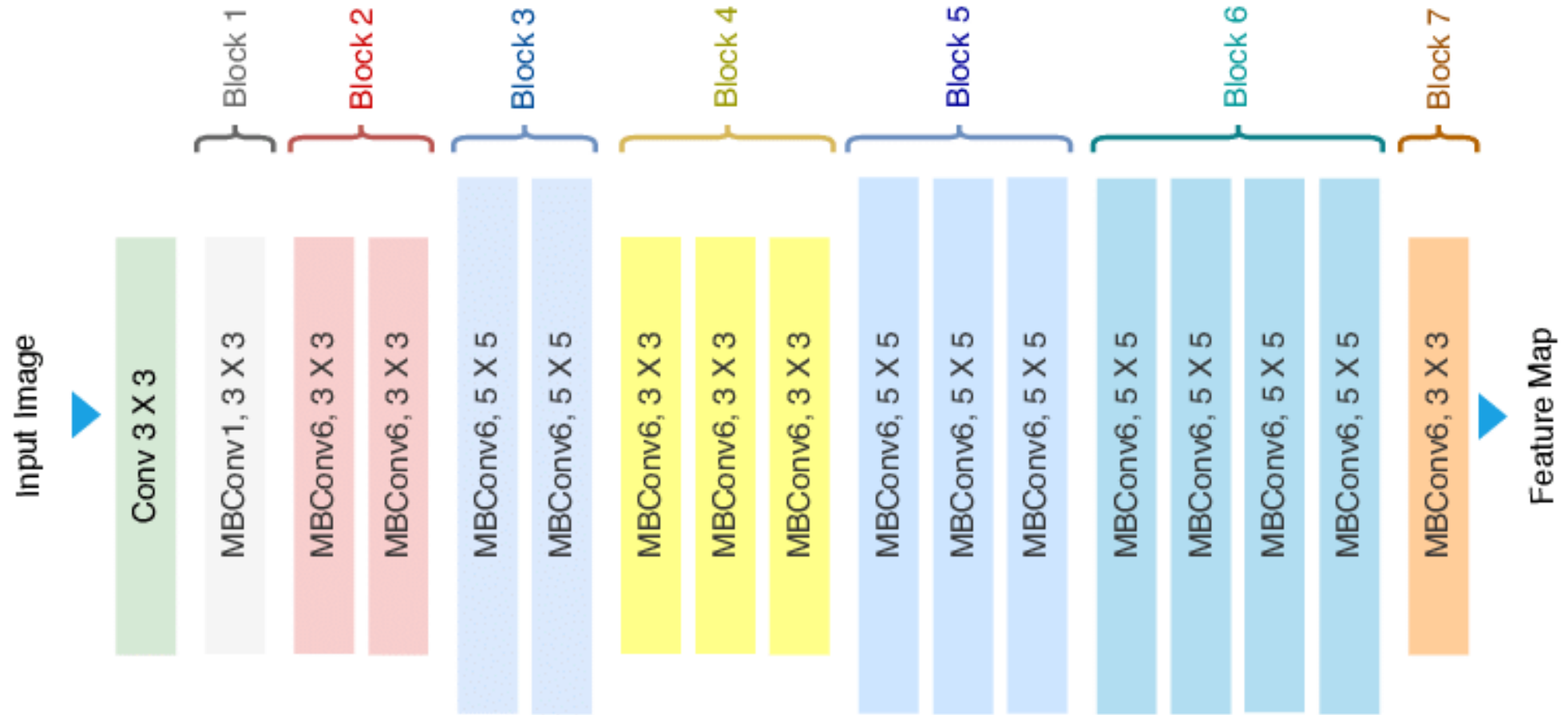
An Analysis of Deep Neural Network Models for Practical Applications, 2017.

MBConv Module



$$\text{Swish}(x) = x \cdot \sigma(x) = x \cdot \frac{1}{1 + e^{-x}}$$

EfficientNet Architecture



Comparison with EfficientNet

